



9/1/2026

N + - Network Fundamentals & Building Networks.

IP routing and routing protocols.

TASK 1.

- ❖ Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

➤ Network Diagram:

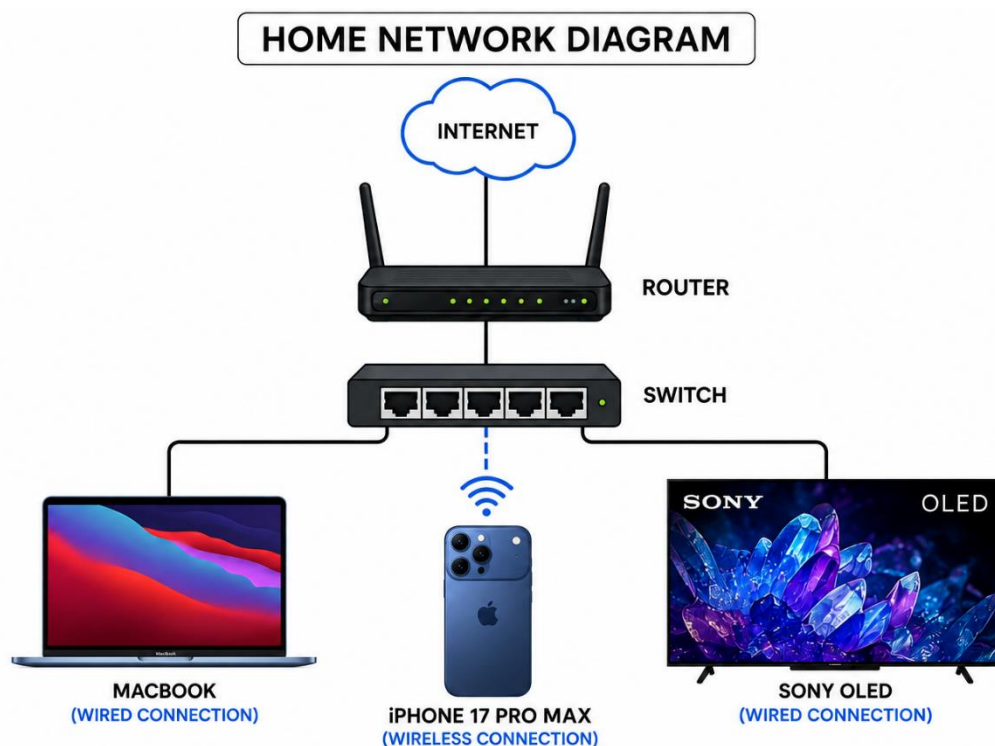


Figure : Network Diagram Showing Router, Switch and Connected Devices.

⌘ IP Addressing:

- **Router LAN Interface:** 192.168.1.1
- **MacBook:** 192.168.1.11
- **iPhone 17 Pro Max:** 192.168.1.10
- **Sony OLED TV:** 192.168.1.12
- **Router WAN Interface:** 203.0.113.1

TASK 2.

- ❖ **Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made, Use made-up IP addresses and show the routing table entries that would be checked at each hop.**

➤ **Manual IP Routing:**

⌘ **Scenario:**

My laptop has IP address 192.168.1.10/24, while my friend's computer has IP address 192.168.2.20/24. Since they are on different subnets, the packet must pass through a router.

⌘ **Step 1: Packet leaves the laptop**

The laptop checks the destination 192.168.2.20. It sees that the destination is outside its own subnet, so it sends the packet to its default gateway, 192.168.1.1.

Destination Network	Next Hop	Interface
192.168.1.0/24	Directly Connected	LAN
192.168.2.0/24	10.0.0.2	WAN

⌘ **Step 2: Router 1 makes the routing decision**

Router 1 checks its routing table and finds 192.168.2.0/24 as the matching destination network. It forwards the packet to the next router at 10.0.0.2.

Destination Network	Next Hop	Interface
10.0.0.0/30	Directly Connected	WAN
192.168.2.0/24	Directly Connected	LAN

⌘ Step 3: Packet reaches the friend's computer

Router 2 identifies 192.168.2.0/24 as directly connected and forwards the packet to 192.168.2.20.

Packet Path: Laptop (192.168.1.10) → Router 1 (192.168.1.1) → Router 2 (10.0.0.2) → Friend's PC (192.168.2.20)

TASK 3.

❖ **Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.**

➤ **Routing Table for Delivery Network:**

Branch / City	Network Address	Subnet Mask	Next Hop	Interface
Mumbai	192.168.10.0	255.255.255.0	10.0.0.2	GigabitEthernet0/0
Delhi	192.168.20.0	255.255.255.0	10.0.0.3	GigabitEthernet0/1
Bengaluru	192.168.30.0	255.255.255.0	10.0.0.4	GigabitEthernet0/2

TASK 4.

❖ **Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).**

➤ **Static Routing vs Dynamic Routing:**

Routing Type	Pros	Cons
Static Routing	1. Simple to configure for small networks. 2. Does not use routing protocol bandwidth. 3. Provides predictable and controlled paths.	1. Requires manual changes when the network changes. 2. Not suitable for large networks. 3. A failed route may need manual correction.
Dynamic Routing (RIP/OSPF)	1. Automatically finds and updates routes. 2. Adapts to network failures and changes. 3. Suitable for large networks, such as food delivery services handling traffic between many servers.	1. Uses CPU, memory, and network resources. 2. More complex to configure and manage. 3. Routing updates can create additional network traffic.

TASK 5.

- ❖ Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does, Paste the AI-generated config and annotate each part with a short description.

➤ OSPF Routing Configuration:

⌘ Sample OSPF Configuration between Two Routers

🔗 Router 1

Router1> enable

Router1# configure terminal

Router1(config)# router ospf 1

Router1(config-router)# router-id 1.1.1.1

Router1(config-router)# network 192.168.1.0 0.0.0.255 area 0

Router1(config-router)# network 10.0.0.0 0.0.0.3 area 0

Router1(config-router)# end

🔗 Annotations:

enable → Enters privileged mode.

configure terminal → Opens global configuration mode.

router ospf 1 → Starts an OSPF routing process.

router-id 1.1.1.1 → Gives Router 1 a unique OSPF identity.

network 192.168.1.0 0.0.0.255 area 0 → Enables OSPF on the 192.168.1.0/24 network.

network 10.0.0.0 0.0.0.3 area 0 → Enables OSPF on the link between the two routers.

end → Exits configuration mode.

↗ **Router 2**

Router2> enable

Router2# configure terminal

Router2(config)# router ospf 1

Router2(config-router)# router-id 2.2.2.2

Router2(config-router)# network 192.168.2.0 0.0.0.255 area 0

Router2(config-router)# network 10.0.0.0 0.0.0.3 area 0

Router2(config-router)# end

↗ **Annotations:**

enable → Enters privileged mode.

configure terminal → Opens global configuration mode.

router ospf 1 → Starts an OSPF routing process.

router-id 2.2.2.2 → Gives Router 2 a unique OSPF identity.

network 192.168.2.0 0.0.0.255 area 0 → Enables OSPF on the 192.168.2.0/24 network.

network 10.0.0.0 0.0.0.3 area 0 → Enables OSPF on the link between the two routers.

end → Exits configuration mode.

